AI TOOL DEVELOPMENT CHALLENGE 2026

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FinInclusion AI

AI-Powered Financial Stability Predictor
for Informal Workers in South Asia

PROJECT OVERVIEW DOCUMENT

Submitted to: AI Tool Development Challenge 2026
Domain: Financial Inclusion | Region: South Asia
SDG Alignment: Goals 1, 8 & 10
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Project Title:	FinInclusion AI — Financial Stability Predictor for Informal Workers
Team Name:	Team FinInclusion
Category:	Machine Learning | Predictive Analytics | Social Impact AI
Tech Stack:	Python | Scikit-learn | Streamlit | Pandas | Matplotlib

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Project link --→ https://github.com/Bhagyesh-CodeForge/finclusion-ai

1. Problem Statement & SDG Alignment

1.1 The Problem of Financial Exclusion

Across South Asia, an estimated 300–400 million workers operate in the informal economy — street vendors, daily wage labourers, gig delivery workers, auto-rickshaw drivers, and small shopkeepers. Despite their significant contribution to national GDP, these workers are systematically excluded from formal financial systems for one critical reason: they have no traditional financial records.

Without payslips, bank statements, or credit history, they cannot access:

- Formal bank loans or micro-credit facilities
- Government welfare and subsidy programs
- Insurance products or emergency financial safety nets
- Digital payment infrastructure tied to formal identity

The Core Gap: Traditional credit scoring systems require formal employment records, bank statements, and tax filings. Informal workers have none of these — not because they are financially irresponsible, but because their economic activity is invisible to formal systems.

1.2 SDG Alignment

This project directly addresses three United Nations Sustainable Development Goals:

SDG 1	**No Poverty — End poverty in all its forms everywhere**

Financial exclusion traps informal workers in cycles of poverty. Without access to credit or welfare, a single financial shock — illness, accident, or crop failure — can be devastating. FinInclusion AI provides an alternative pathway to financial support.

SDG 8	**Decent Work & Economic Growth — Promote sustained, inclusive and sustainable economic growth**

By enabling micro-credit access for informal workers, the platform supports entrepreneurship, business expansion, and economic mobility — directly contributing to inclusive economic growth in underserved communities.

SDG 10	**Reduced Inequalities — Reduce inequality within and among countries**

The AI model uses alternative indicators to level the playing field — giving informal workers the same quality of financial assessment as formally employed citizens, regardless of their lack of traditional records.

2. Proposed Solution & AI Approach

2.1 Solution Overview

FinInclusion AI is a machine learning-powered platform that analyses alternative economic indicators to assess financial stability and generate actionable recommendations for informal workers — without requiring any formal financial records.

Core Innovation: Instead of asking 'What is your bank balance?', the system asks 'How consistently do you earn? How often do you transact? How digitally active are you?' — building a financial profile from behaviour rather than formal records.

2.2 AI & Machine Learning Approach

The system evaluates two state-of-the-art ensemble models and automatically selects the best performer:

Model	Description & Characteristics
Random Forest Classifier	Ensemble of 100 decision trees. Handles non-linear relationships well. Balanced class weights to account for unequal label distribution. Achieved 89.9% accuracy on test data.
Gradient Boosting Classifier	Sequential boosting ensemble with 100 estimators and learning rate 0.1. Superior at capturing complex feature interactions. Auto-selected with 90.5% test accuracy.

2.3 Input Features (Alternative Indicators)

The model learns from the following alternative economic signals — all collectible without formal records:

Feature	What It Measures	Why It Matters
Daily Income (₹)	Average earnings per working day	Core indicator of earning capacity
Work Days / Month	Number of active working days	Measures employment consistency
Monthly Transactions	Number of financial transactions	Proxy for economic activity level
Savings Ratio	Fraction of income saved	Reflects financial discipline
Digital Payment Ratio	Share of digital payments	Indicates financial inclusion level
Loan History	Existing credit exposure (binary)	Signals prior credit engagement
Income Consistency	Ratio of min to max income	Measures income stability
Monthly Income	Total estimated monthly earning	Comprehensive income signal

2.4 Output & Recommendations

For each worker assessed, the system produces four actionable outputs:

Output	Details
Financial Stability Label	Classified as Stable, Moderate, or Unstable based on ML prediction
Alternative Credit Score	Scaled 300–850 (matching traditional credit score ranges for familiarity)
Risk Category	Low / Medium / High — derived from credit score bands
Support Program Recommendation	Micro-Investment Scheme / Micro-Credit + Skills Training / Government Welfare

3. System Architecture & Workflow

3.1 Five-Layer Architecture

The platform is structured into five logical layers, each with a distinct responsibility:

Layer	Components & Responsibility
Layer 1 — Data Collection	Collects worker-provided inputs via Streamlit UI: income, occupation, transaction frequency, savings ratio, digital payment ratio, and loan history.
Layer 2 — Data Processing	Cleans raw inputs, engineers derived features (income_consistency, monthly_income), applies Label Encoding to categorical variables, and normalises numerics with MinMaxScaler.
Layer 3 — AI Model	Gradient Boosting Classifier trained on 5,000 synthetic worker profiles. Outputs class probabilities across three stability classes.
Layer 4 — Decision Support	Converts raw probabilities into credit score (300–850), risk category, and maps outcome to recommended welfare or micro-credit program.
Layer 5 — Interface	Streamlit web application with interactive input sliders, credit score gauge, feature contribution waterfall chart, and plain-English explanation.

3.2 End-to-End Prediction Workflow

A single prediction follows this sequence from user input to recommendation:

1. Worker enters profile data via the Streamlit sidebar (7 input fields).
2. Preprocessing pipeline engineers income_consistency and monthly_income, encodes occupation, and applies MinMaxScaler normalisation.
3. Feature vector is passed to the trained GradientBoostingClassifier (loaded from financial_model.pkl).
4. Model outputs probability scores for each of the three stability classes (Unstable / Moderate / Stable).
5. Credit score is computed as a weighted probability sum: P(Stable)×1.0 + P(Moderate)×0.5, scaled to 300–850.
6. Risk category is assigned based on credit score bands: Low (>=700), Medium (500–699), High (<500).
7. Factor Contribution Analysis computes per-feature impact using feature importance × deviation from average worker.
8. Results displayed: stability label, credit score gauge, waterfall chart, probability breakdown, and support program recommendation.

3.3 Project File Structure

```
finclusion/
    data/raw/informal_workers.csv          5,000 synthetic worker records
    data/processed/processed_workers.csv   Cleaned, encoded, normalised dataset
    models/financial_model.pkl             Trained GradientBoosting model
    utils/preprocess.py                    Full preprocessing pipeline
    utils/model.py                         ML training, evaluation & selection
    app/main.py                            Streamlit web prototype
    requirements.txt                       All Python dependencies
```

4. Expected Impact & Use Cases

4.1 Direct Beneficiaries

The platform is designed to serve multiple stakeholder groups simultaneously:

Stakeholder	How FinInclusion AI Helps Them
Informal Workers	Receive an alternative credit score and clear recommendation for micro-credit or welfare programs — without needing formal financial records.
Microfinance Institutions (MFIs)	Automate and standardise loan eligibility assessment for informal applicants, reducing manual review time and improving decision consistency.
Government Welfare Agencies	Identify workers most likely to qualify for welfare programs (e.g., PM Jan Dhan Yojana) using AI screening rather than manual paperwork.
NGOs & Field Workers	Use the tool on a tablet or phone to assess workers in the field and connect them to appropriate financial services on the spot.
Policymakers & Researchers	Aggregate anonymised data to understand financial vulnerability patterns across regions, occupations, and income levels.

4.2 Real-World Use Cases

- **Use Case 1:** A street vendor in Mumbai applies for a ₹15,000 micro-loan. The MFI uses FinInclusion AI to assess eligibility in under 60 seconds — no bank statement required.
- **Use Case 2:** A government field officer screens 50 daily wage labourers in a rural village for welfare eligibility using the mobile prototype, completing in one afternoon what previously took weeks of paperwork.
- **Use Case 3:** An NGO running a financial literacy program uses the credit score output to track improvement in workers' financial stability over 6 months of training.
- **Use Case 4:** A researcher analyses anonymised model outputs across 10,000 workers to identify which occupations are most financially vulnerable in post-monsoon months.

5. Scalability & Deployment Considerations

5.1 Scaling the Model

The current prototype is built for rapid demonstration. The following roadmap outlines how it scales to production:

Phase	Scalability Action
Phase 1 — POC (Current)	Synthetic dataset of 5,000 records. Streamlit prototype. Single-machine deployment.
Phase 2 — Pilot	Partner with 1–2 MFIs or NGOs in India/Bangladesh to collect real worker data. Retrain model on 50,000+ real records. Deploy on cloud (AWS/GCP).
Phase 3 — Regional Scale	Multilingual UI (Hindi, Urdu, Bengali, Tamil). REST API so MFIs can integrate directly into their loan management systems. Mobile-first PWA interface.
Phase 4 — National Scale	Federated learning across partner organisations to improve model without centralising sensitive data. Integration with Aadhaar / DigiLocker for verified data inputs.

5.2 Deployment Architecture

- Backend: FastAPI or Flask REST API exposing the prediction endpoint
- Model serving: Joblib-serialised model loaded at startup, cached in memory
- Database: PostgreSQL for storing anonymised assessment records
- Frontend: Streamlit (prototype) → React PWA (production)
- Cloud: Docker container deployed on AWS EC2 or Google Cloud Run
- Security: All personally identifiable information encrypted at rest and in transit

5.3 Data Privacy & Ethics

The platform is designed with responsible AI principles at its core:

- No raw personal data is stored after prediction — only anonymised aggregate statistics
- The Factor Contribution Analysis provides full transparency into why a score was given
- Workers can request re-assessment after improving their financial behaviour
- The model is periodically audited for bias across occupation groups and income levels
- Consent-first design: workers explicitly agree to data use before assessment

6. KPIs & Performance Indicators

6.1 Model Performance KPIs

The following metrics are tracked after every model training run:

KPI	Current Value (POC)	Target (Production)
Model Accuracy	90.5%	>= 88% on real-world data
Precision — Stable class	89%	>= 85%
Recall — Unstable class	92%	>= 90% (critical — must not miss vulnerable workers)
F1 Score (Macro Average)	0.90	>= 0.87
Inference Time per Prediction	< 100ms	< 200ms at scale
Model Retraining Frequency	On demand	Quarterly with new real-world data

6.2 Business & Social Impact KPIs

Beyond model accuracy, the platform tracks real-world outcomes:

KPI	Measurement Method	12-Month Target
Workers Assessed	Count of unique predictions made	10,000 workers
Micro-credit Approvals Enabled	Partner MFI approval rate for AI-recommended workers	2,000 loan approvals
Welfare Program Connections	Workers successfully enrolled in government schemes	1,500 workers enrolled
False Denial Rate	% of truly eligible workers incorrectly classified as Unstable	< 8%
User Satisfaction Score	Field officer feedback survey	>= 4.0 / 5.0
Average Assessment Time	Time from form open to result display	< 2 minutes

6.3 Explainability & Fairness KPIs

- Bias Audit Score: Prediction accuracy must not vary by more than 5% across occupation groups
- Explainability Coverage: 100% of predictions must include a Factor Contribution Analysis chart
- Appeal Rate: Target < 10% of workers disputing their assessment (measures model trust)
- Demographic Parity: Credit score distributions must be statistically similar across gender and region

7. Conclusion

FinInclusion AI demonstrates that artificial intelligence can be a powerful tool for social equity — not just economic efficiency. By replacing traditional credit history with alternative behavioural indicators, the platform gives a financial voice to the hundreds of millions of informal workers who have been systematically invisible to formal financial systems.

The POC achieves 90.5% accuracy on synthetic data and produces a fully functional Streamlit prototype with explainable predictions. With real-world data partnerships and the deployment roadmap outlined above, this platform has the potential to directly improve the financial lives of millions across South Asia.

> "*Financial inclusion is not just about access to money.*"
>
> *It is about access to dignity, opportunity, and economic security.*